

Detector-Conditioned Observability of Interference Under Coarse-Grained Measurement

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Abstract

Interference visibility is commonly treated as an intrinsic property of wave superposition. In practical systems, however, visibility is inferred through detector sampling and noise-affected measurements.

We present a computational measurement framework integrating wave propagation, detector discretization, noise modeling, and visibility estimation. Building on an initial measurement stack implementation, we extend the framework to quantify the detector resolution required to sustain observable interference.

We find an empirical scaling: $B_{\min} \propto d^{-0.77}$, which deviates from ideal Nyquist scaling (d^{-1}). This deviation arises from a trade-off between spatial resolution and noise suppression, resulting in a stochastic observability boundary.

1. Introduction

Interference is a fundamental signature of coherence in wave systems. In ideal treatments, fringe visibility depends on system geometry and wave properties.

However, real measurement systems introduce finite detector resolution, shot noise, and read noise, affecting observed visibility.

2. Measurement Stack Foundation

The computational pipeline is: field $\rightarrow$ propagation $\rightarrow$ detector $\rightarrow$ noise $\rightarrow$ measurement.

Key components include Fresnel propagation, detector sampling, noise modeling, and interference reconstruction.

3. Extended Framework

We introduce pixel binning, visibility extraction, threshold-based observability, and statistical averaging.

Updated pipeline: field $\rightarrow$ propagation $\rightarrow$ detector $\rightarrow$ binning $\rightarrow$ noise $\rightarrow$ visibility $\rightarrow$ decision. Criterion: $V \geq 0.75$.

4. Experiment

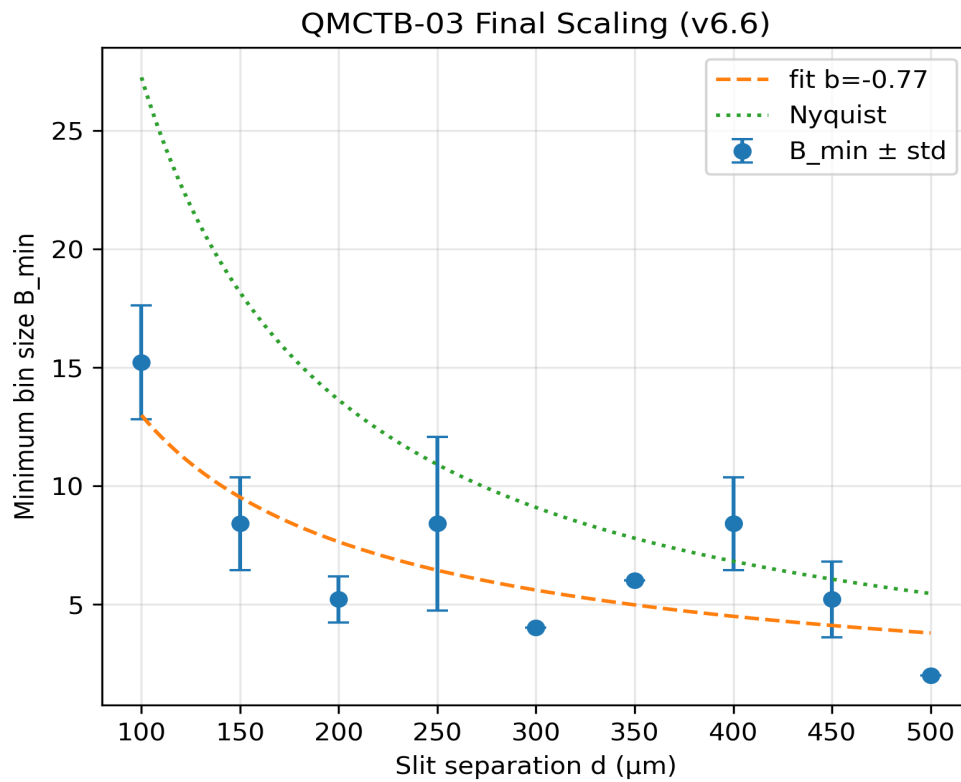
We simulate a double-slit system with slit separation $d \in [100, 500] \mu\text{m}$.

Outputs include mean B_{min} and standard deviation across multiple runs.

5. Results

We observe scaling: $B_{\text{min}} \propto d^{-0.77}$.

Figure 1: Scaling of Detector Resolution vs Slit Separation



Minimum detector bin size (B_{min}) as a function of slit separation (d). Dashed curve shows fitted scaling, dotted curve shows Nyquist prediction.

6. Interpretation

High resolution preserves fringes but amplifies noise, while low resolution suppresses noise but blurs fringes. This creates a resolution–noise trade-off.

7. Core Insight

Interference visibility is not purely intrinsic; it is conditioned by detector resolution, noise statistics, and measurement criteria.

8. Reproducibility

Code available at: <https://github.com/srikarr20/quantum-measurement-stack-demo>

9. Conclusion

Interference observability is probabilistic under noise and depends on detector characteristics.